Task 3: Dataset Understanding and Descriptive Analysis

Project: Predictive Anemia Screening and Healthcare Resource Planning Using Full Blood Count Data in Sri Lankan Healthcare Institutions

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3.1 Dataset Description

The working dataset (fbc_anemia_clean.csv) was constructed by our team from the publicly available Sri Lankan Full Blood Count (FBC) dataset on Zenodo (222,042 long-format lab-parameter records from four anonymized healthcare institutions, 2014–2023). The original data had no case-level identification - each row was a single lab parameter reading, not a patient record - so we engineered a **case-level (wide-format) dataset**, pivoting the six hematology parameters common to all four institutions (HB, MCH, MCHC, MCV, PCV, RBC) into one row per patient test event, and derived a WHO-threshold-based anemia classification for each case.

Property	**Value**
Rows (unique cases)	27,034
Columns	16
Institutions	4 (Institution A - D)
Date range	21 Apr 2014 – 11 Aug 2023
Unit of analysis	One patient FBC test event

Organizational perspective: transforming raw lab-parameter logs into a case-level dataset is itself the first deliverable of this engagement - the client organization did not previously have a structure that let them ask "how many patients were anemic?" directly; they only had individual test-value logs. This restructuring is what makes every subsequent statistical question in this report answerable.

3.2 Variable Descriptions

Variable	Type	Description
case_id	Identifier	Unique key per test event (engineered during pivot)
institution	Categorical (4 levels)	Anonymized healthcare institution
sex	Categorical (2 levels)	Female / Male
age_at_test	Numeric	Patient age in years at time of test
birth_year	Numeric	Patient birth year (source field for age)
bill_date	Date	Date of the test/billing event
HB	Numeric (g/dL)	Hemoglobin concentration
MCH	Numeric (pg)	Mean Corpuscular Hemoglobin
MCHC	Numeric (g/dL)	Mean Corpuscular Hemoglobin Concentration
MCV	Numeric (fL)	Mean Corpuscular Volume
PCV	Numeric (%)	Packed Cell Volume (hematocrit)
RBC	Numeric ($\times 10^{12}$/L)	Red Blood Cell count

Variable	Type	Description
who_hb_threshold	Numeric (g/dL)	Age/sex-specific WHO HB cut-off applied to this case
anemia	Categorical (Y/N)	Derived: HB below the applicable WHO threshold
anemia_severity	Ordinal	Derived: Normal / Mild / Moderate / Severe (WHO severity bands)
age_group	Categorical (8 levels)	Engineered age band for subgroup analysis (added by our team for this task)

Organizational perspective: the six retained parameters are exactly the "red-cell panel" already produced by any standard automated hematology analyzer used across the client's institutions, meaning any tool built on these variables requires **no new data collection or equipment investment**.

3.3 Data Quality Assessment

3.3.1 Missing Value Analysis

Every one of the 16 variables has **zero missing values** in the final dataset (0.00% missingness across the board). This is not incidental - it reflects the upstream cleaning already performed by our team: cases with missing HB or age were dropped during the original long-to-wide pivot, and only cases with a complete red-cell panel were retained.

Organizational perspective: while zero missingness is convenient for modelling, it is worth being transparent with the client that this reflects a *complete-case* dataset - cases with partial FBC panels (e.g. a rushed or interrupted test) were excluded, not imputed. If the true rate of incomplete testing in the client's institutions is high, this

dataset may present a slightly more "complete picture" than day-to-day operational reality, which we flag as a limitation for any operational deployment.

3.3.2 Outlier Detection

Using the standard IQR method (values beyond 1.5×IQR from Q1/Q3):

Parameter	Lower fence	Upper fence	Outliers (n)	Outliers (%)
HB	7.5	17.1	295	1.09%
MCH	23.2	34.4	1,327	4.91%
MCHC	29.8	36.6	863	3.19%
MCV	71.2	102.5	1,235	4.57%
PCV	23.1	51.1	279	1.03%
RBC	2.76	5.88	391	1.45%

Critically, we cross-checked these *statistical* outliers against **known clinically plausible ranges** (e.g. HB 3–25 g/dL, MCHC 25–40 g/dL - MCHC is a tightly biologically-constrained ratio). All values across all six parameters fall within physiologically plausible bounds - the extremes are consistent with genuine clinical conditions (e.g. severe anemia, macrocytosis) rather than data entry errors.

Organizational perspective: these "outliers" are not noise to be removed - they represent the sickest patients in the dataset, who are precisely the population a screening tool most needs to correctly identify. We recommend **retaining all statistical outliers** in the modelling dataset (Task 5), while continuing to monitor for genuine data-entry artefacts.

3.3.3 Resolved Data Quality Issue: Implausible Ages

During inspection, **12 cases (0.044% of the pre-cleaning dataset)** were found with implausible ages - up to 858 years - traced to birth-year entry errors (e.g. birth_year = 1165) inherited from the original health-management information system records. These 12 rows were **dropped** (not imputed or capped), as fabricating a plausible age would introduce more bias than the negligible loss of statistical power from removing 0.044% of records.

Organizational perspective: this is a useful, low-cost finding to report back to the client - it points to a specific, fixable data-entry validation gap in the source HMIS (no upper bound check on derived age), and is the kind of small operational recommendation a consultancy engagement is expected to surface, distinct from the main statistical analysis.

3.4 Descriptive Statistics

Parameter	Mean	SD	Median	Min	Max	Q1	Q3
Age (years)	38.1	22.2	37.0	1	99	22	56
HB (g/dL)	12.3	1.86	12.2	3.2	20.9	11.1	13.5
MCH (pg)	28.6	2.66	29.0	15.2	42.1	27.4	30.2
MCHC (g/dL)	33.2	1.50	33.2	25.1	39.9	32.3	34.0
MCV (fL)	86.1	7.06	87.1	50.8	126.0	82.8	90.5
PCV (%)	37.1	5.34	36.9	10.0	60.1	33.6	40.6
RBC ($\times 10^{12}$/L)	4.32	0.61	4.30	1.19	7.44	3.93	4.71

All six parameters show **near-zero skew** (|skew| < 0.2), i.e. approximately symmetric distributions.

Categorical breakdown

- **Sex:** 57.7% Female (n=15,597), 42.3% Male (n=11,437)
- **Institution:** Institution_B dominates the sample (65.3%, n=17,652); Institution_A is smallest (2.6%, n=710)
- **Anemia:** 43.6% of all cases (n=11,793) are classified as anemic by WHO threshold
- **Severity (among all cases):** Normal 57.6%, Mild 23.3%, Moderate 18.0%, Severe 1.1%
- **Age structure:** heavily adult-skewed - 34.6% aged 18-39, 24.8% aged 40-59; children (0-11) and elderly (80+) are comparatively under-represented (≈15% and 2.4% respectively)

Organizational perspective: an overall anemia prevalence of 43.6% is high in absolute terms and broadly consistent with WHO's characterization of anemia as a public health problem when prevalence exceeds ~20% in a population - reinforcing the priority of this issue for the client. However, this raw pooled figure is misleading on its own, as the next sections show substantial variation by subgroup.

3.5 Key Visualizations and Insights

3.5.1 Hemoglobin distribution

The overall HB distribution is roughly unimodal and only slightly right-skewed, with the anemic/non-anemic split cutting through the middle of a continuous distribution (not two separate populations) - confirming that anemia status, as defined here, is a **thresholder continuous risk**, not a naturally separate clinical group. This matters methodologically: it supports treating HB as a continuous outcome in some analyses (Task 4) while also justifying the binary/ordinal anemia classification for others (Task 4, 5).

3.5.2 Anemia prevalence by sex - and its reversal with age

The single most important descriptive finding in this dataset is the **interaction between age and sex** in anemia risk:

- In the **18–39 age group**, women are anemic at **51.6%** versus just **9.5%** of men - a >5x gap, consistent with known menstrual/reproductive-age iron loss.
- This gap **narrows through middle age** and **reverses by 60-79** (69.8% women vs. 57.4% men) and **80+** (73.7% women vs. **80.5% men**), where elderly men show higher anemia prevalence than elderly women.

Organizational perspective: this is not a generic finding - it is directly actionable. A single, undifferentiated "anemia screening threshold" or awareness campaign targeted at "women" would badly under-serve elderly male patients, who show the highest prevalence of any subgroup in the dataset. This single chart is strong evidence for **age-and-sex-stratified**, not blanket, resource allocation and screening policy.

3.5.3 Anemia prevalence by institution

Prevalence ranges from 41.6% (Institution_D) to 57.3% (Institution_C), with Institution_A and B closer to the pooled average (45.5%, 42.1%). Institution_C stands out as notably higher.

Organizational perspective: this raises a genuine question for Task 4 (is this difference statistically significant, or attributable to Institution_C's specific patient population - e.g. if it serves an older or more rural catchment?) - we flag institution as worth formal hypothesis testing, while also noting the severe sample size imbalance below as a caveat on how confidently any institution-level comparison can be interpreted.

3.5.4 Sample size imbalance across institutions

Institution_B alone accounts for 65.3% of all cases; Institution_A only 2.6%. This imbalance was a known feature of the source Zenodo dataset and is preserved here.

Organizational perspective: any institution-level comparison (including 3.5.3 above) carries a caveat - estimates for Institution_A rest on a much smaller sample (n=710) than Institution_B (n=17,652), so its prevalence estimate is inherently less precise. We will account for this explicitly (e.g. with confidence intervals, not just point estimates) in Task 4, and it is part of the justification for **excluding institution as a predictor** in the generalizable screening model in Task 5.

3.5.5 Anemia severity distribution

Most anemic cases are Mild (23.3% of all cases) or Moderate (18.0%); Severe anemia is comparatively rare (1.1%, n=294) but clinically the most urgent.

Organizational perspective: because severe cases are rare, any predictive model or resource-planning tool must be designed not to let this small-but-critical group get statistically "washed out" by the much larger mild/moderate categories - a direct input into how we frame the Task 5 model's evaluation criteria (e.g., recall on severe cases may matter more than overall accuracy).

3.5.6 Correlation between FBC parameters

HB and PCV are near-collinear ($r = 0.96$); MCH and MCV are highly correlated ($r = 0.88$); RBC correlates strongly with both HB (0.78) and PCV (0.82). MCHC is comparatively independent of the others (all $|r| \leq 0.5$).

Organizational perspective: this multicollinearity is expected - these parameters are mathematically and physiologically linked (PCV and RBC are components of how HB concentration arises). It is important evidence for two later tasks: it motivates *caution* in Task 5 if multiple red-cell parameters are used together as predictors (variance inflation), and it is direct, dataset-specific justification for the Task 7 discussion of whether Principal Component Analysis is worth applying here.

3.5.7 Age distribution

The patient population is adult-heavy, peaking in the 20s–50s, with comparatively few children under 5 or adults over 80 relatives to typical population pyramids - likely reflecting which patient groups are referred for FBC testing in these institutions (e.g., routine adult check-ups, chronic disease monitoring) rather than the general population structure.

Organizational perspective: this is an important scope caveat for the client - this dataset is not a population-representative health survey; it reflects **who was tested**, which is shaped by existing referral patterns, not random sampling. Any conclusions about "population anemia burden" should be read as burden **among tested patients**, and we recommend this distinction be made explicit in any communication of findings to non-technical stakeholders.

Summary of Task 3 Findings for the Consultancy Narrative

1. Dataset is clean, complete, and analysis-ready after removing 12 data-entry-error records (0.044%).
2. Overall anemia prevalence (43.6%) is high enough to be a genuine public health concern.
3. **Age × sex interaction is the standout finding** - reproductive-age women and elderly men are the two highest-risk groups, for different underlying reasons, requiring differentiated policy responses.
4. Institution-level differences exist but must be interpreted cautiously given sample imbalance.
5. Severe cases are rare but clinically critical - evaluation of any future model must protect against ignoring this subgroup.
6. Strong multicollinearity among red-cell parameters is expected and has direct implications for Tasks 5 and 7.